DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #01E: PATHS & SUBPATHS (PART 5 — GRANULAR AUDIT OF GRAND DAO WAVE 1A: `ASTRO_DIVINATION`, `BUDDHA_DHARMA`, `VOID_SPACE`, `CELESTIAL_WEATHER`, `DRAGON_VEIN_GEOMANCY`, `GRAVITY_DAO`, `CHRONO_EPOCH`, `PRIMORDIAL_CHAOS` & ALL 48 SUBPATHS)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Series Position (`01C–01H` Granular 43-Path / 286-Subpath Audit)**:
  - `01A–01B`: Shared Macro Lifecycle, Path Library, Retraining & Stage Equivalence Math.
  - `01C`: Group 1 — 5 Founding Pillars (`XIAN`, `BODY`, `GOD`, `DEMON`, `HEART_SOUL` — 65 Subpaths).
  - `01D`: Group 2 — 7 Specialist Expansion Paths (`MECH_PUPPET`..`STRATEGY_GOVERNANCE` — 67 Subpaths).
  - **`01E` (This Document)**: Group 3 — **8 Cosmic, Spacetime, Weather, Dharma & Geomancy Grand Dao Paths (`ASTRO_DIVINATION` [6], `BUDDHA_DHARMA` [6], `VOID_SPACE` [6], `CELESTIAL_WEATHER` [6], `DRAGON_VEIN_GEOMANCY` [6], `GRAVITY_DAO` [6], `CHRONO_EPOCH` [6], `PRIMORDIAL_CHAOS` [6] — 48 Subpaths total)**.

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order & Strict Stone Conservation**: Preserve exact `<script>` order in `index.html`. Every stone spent by `anchorDragonVeinNode` or sect rituals must route through `ECO.pay(W, cost, reason)` so `W.world.resourcePool.stones` stays strictly balanced (`Debit === Credit`).

3. **Replace Monthly Cron Hacks With Real Situational NPC Autonomy**: Do **not** drive path mechanics via blind `SIM.monthly` scripts that teleport actions while NPCs are jailed or away (`n.away`). Drive them via authentic `AI.ACTS` stations, pre-battle tactical hooks, and officer duties.

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1. `[GDAO-STAT-01]` Systemic Stat-Engine Bugs Shared by All 31 Grand Dao Paths (`js/stats.js`)

Inspecting `js/stats.js` (`compute` lines 250–284 vs `getBaseFast` lines 459–518) reveals **three systemic bugs** affecting every path in `01E–01H`:

1. **`subpathTraits` Milestone Lockout (`stats.js:503`)**:

```javascript
const curOrd = (c.track && c.track.ordinal) ? c.track.ordinal : ((c.path === 'BODY' || c.path === 'GOD') && CULT.normalizedStage ? Math.floor(CULT.normalizedStage(n)) : c.stage);
```

   - Every Grand Dao subpath defines 3 milestone `subpathTraits` at **`reqRealm: 3, 6, 9`** (`v200` line 121).
   - **Case A (`c.track` initialized at `ordinal = 1` on recruited/hydrated elders)**: `curOrd` evaluates to **`1`**, completely ignoring `c.stage = 7` or `8`! A Stage-8 Grandmaster receives **zero** `subpathTraits` in combat (`ST.get`), while `ST.explain` (`compute` line 273) calls `CULT.GRAND_DAO_MECHANICS.getActiveTraits(n)` which uses a different stage check—creating another **UI vs Combat split-brain**!
   - **Case B (`!c.track` on 8-stage paths)**: Because `normalizedStage` is only checked for `'BODY'` and `'GOD'`, all 8-stage Grand Dao paths cap at `curOrd = 8`, **never reaching `reqRealm: 9`**!
   - **Fix**: Have **both** `compute` and `getBaseFast` call `CULT.GRAND_DAO_MECHANICS.getActiveTraits(n)`, where `curOrd = Math.max(c.track ? c.track.ordinal : 0, Math.round(CULT.normalizedStage(n)))`!

2. **Weak Generic Mantra Fallback (`stats.js:261, 470`)**:

   - Only the first 12 paths have explicit Mantra stat formulas in `stats.js`. All 31 Grand Dao paths fall into `else { mdef +0.04*tier, hp +0.03*tier, matk +0.02*tier }`—granting **`+0% atk`** to physical/hybrid paths like

GRAVITY_DAO (HEAVY_STAR_CRUSHER), BUDDHA_DHARMA (VAJRA_WRATH_GUARDIAN), and PRIMORDIAL_CHAOS (ABYSSAL_CHAOS_HALBERD), and only half the matk scaling of XIAN (0.02 vs 0.04)!
- **Fix**: Derive Mantra stat scaling dynamically from PATHW[c.path] (statWeights)!

3. **Duplicate Uncalled Helper Methods (M1 Drift Between cultivation.js & combat.js)**:

    - CULT.GRAND_DAO_MECHANICS.triggerTemporalRewind and transmuteChaosElement in js/cultivation.js have **zero callers** outside v200 unit tests, while COMBAT.COMPOSABLE_ENGINE runs separate inline logic (TEMPORAL_SAND_REWIND, CHAOS_ELEMENT_SHATTER) in js/combat.js. Route combat.js through the canonical GRAND_DAO_MECHANICS methods (keeping v200 return signatures intact) so balance changes never diverge!

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2. [PATH-ASTRO-02] Path #13: ASTRO_DIVINATION (วิถีโหราดวงดาวและชะตาลิขิต — 6 Subpaths)

- **Subpaths**: STAR_CHART_PROPHET, POLARIS_FATE_WEAVER, ECLIPSE_DOOM_SEER, METEOR_OMEN_CALLER, QIMEN_EIGHT_GATES_DIVINER, BIG_DIPPER_LONGEVITY_SAGE.
- **Verified Code & Context Defects**:
    1. **Subpath ID Typo in ELEM.syncSpecialAccess (js/cultivation.js)**: Checks 'STAR_CHART_ORACLE' instead of the real ID 'STAR_CHART_PROPHET', failing to grant native ASTRAL elemental access!
    2. **Zero Nocturnal & Celestial Alignment Synergy**: Lore states resource is **"ปราณดาราจักรยามค่ำคืน"** (*Nocturnal Starlight Qi*), yet CULT.rate(n) and STAR_DESTINY gauge gain are identical at midday and midnight, and ASTRO_DIVINATION NPCs show zero preference for night-time star-gazing meditation or SECT.CELESTIAL_ALIGNMENTS (ASTRAL_METEOR_SHOWER, BLOOD_MOON_ECLIPSE)!
    3. **Zero Prophetic Utility in Sect Intelligence, Exploration & Longevity**:
        - STAR_CHART_PROPHET & QIMEN_EIGHT_GATES_DIVINER: Should boost intel/scout early raid detection (+20% warning lead time) and -22% ancient-ruin puzzle/trap risk in explore.js!
        - BIG_DIPPER_LONGEVITY_SAGE (สายกระบวยดาราต่อชะตา): Inspired by the Seven-Star Lamp Longevity Rite, yet grants **zero extra lifeMax** (CULT.lifeMax ignores it!) and cannot extend a dying Elder's lifespan!

- `ECLIPSE_DOOM_SEER`: Should gain `+22%` `STAR_DESTINY` & crit potency during night/`BLOOD_MOON_ECLIPSE`, inflicting `badOmen` (`-crit / -eva`) on enemies, with a small `STAR_DESTINY` backlash if over-casting fate reversals (ฝืนลิขิตฟ้าเกินขีดจำกัด)!

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3. [PATH-BUDDHA-03] Path #14: BUDDHA_DHARMA (วิถีพุทธธรรมและมรรคาวาสนา — 6 Subpaths)

- **Subpaths**: `VAJRA_WRATH_GUARDIAN`, `GOLDEN_LOTUS_CHAN`, `SUTRA_CHANT_ARHAT`, `THOUSAND_PALM_BODHISATTVA`, `RELIC_PAGODA_KEEPER`, `KARMA_FIRE_ASCETIC`.
- **Verified Code & Context Defects**:
  1. **Meditating (`A.cultivate.tick`) Fails to Cleanse `heartDemon` on `BUDDHA_DHARMA` (`js/ai.js`)!**:
     - `HEART_SOUL` and `CRAFT_CREATION` reduce `heartDemon` during meditation, whereas `BUDDHA_DHARMA` (`GOLDEN_LOTUS_CHAN` ฌานบัวทองชำระจิต!) gets **zero** `heartDemon` cleansing in `A.cultivate.tick`!
  2. **Missing Precept Morality & Nonlethal Compassion vs Wrathful Dharma Dichotomy**:
     - Lore explicitly states: **"สะสมบารมีโพธิญาณจากการคุ้มครองสหาย... หากสังหารไร้เหตุผล บารมีธรรมจะเสื่อมถอยและเกิดมารในใจ"**.
     - Currently, a `THOUSAND_PALM_BODHISATTVA` or `GOLDEN_LOTUS_CHAN` monk slaughters low-tier mortals with `0` `BODHI_MERIT` penalty and `0` preference for `VOW_ASCETIC_PRECEPT` / `VOW_NO_KILL`!
     - **Emergent Fix**:
       - Merciful subpaths (`GOLDEN_LOTUS_CHAN`, `SUTRA_CHANT_ARHAT`, `THOUSAND_PALM_BODHISATTVA`, `RELIC_PAGODA_KEEPER`) naturally adopt `VOW_ASCETIC_PRECEPT`, gain `+CULT.gain` & `BODHI_MERIT` from **sparing defeated foes (`yield`), shielding allies, and counseling sectmates' Heart Demons (`SECT.HEART_DEMON.counselHeartDemon`)**, and suffer `heartDemon += 12` if they execute surrendering/virtuous targets!
       - Wrathful Dharma-Protector subpaths (`VAJRA_WRATH_GUARDIAN`, `KARMA_FIRE_ASCETIC`) gain `+20%` righteous smite damage & `BODHI_MERIT` specifically when fighting `DEMON`, `VOID_HERETIC_PACT`, `DEVOURING_FLESH`, `NETHER_GRAVE`, or foes with `sinKarma >= 30`!

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4. [PATH-VOID-04] Path #15: VOID_SPACE (วิถีสุญญตาและมิติว่างเปล่า — 6 Subpaths)

- **Subpaths**: DIMENSIONAL_BLADE_REAPER, VOID_BLINK_STRIDER, POCKET_REALM_ARCHITECT, SINGULARITY_COLLAPSE_MAGE, PHASE_MIRAGE_ASSASSIN, SPATIAL_BARRIER_WARDEN.
- **Verified Code & Context Defects**:
  1. **POCKET_REALM_ARCHITECT (สายสถาปนิกมิติย่อย) Has Zero Spatial Inventory/Shelter Utility**:
     - Building a Pocket Dimension should increase World-Map expedition carryCapacity by **+45%** in EXPED.createParty and let an expedition party shelter inside a spatial pocket once per trip to avoid catastrophic storm/ambush attrition! Currently 0% wired!
  2. **Missing Ranged Projectile Absorption (VOID_POCKET_DEFENSE & SPATIAL_BARRIER_WARDEN)**:
     - Lore states: **"ดูดกลืนโพรเจกไทล์ลงสู่มิติย่อย"**. In COMBAT.strike, attacks from LONG_BALLISTIC weapons (BOW, NEEDLE, BALLISTA) against SPATIAL_BARRIER_WARDEN (or allies behind them while voidPhase / VOID_POCKET_DEFENSE is active) are not specifically intercepted or absorbed into VOID_RIFT gauge!
  3. **Missing VOID_RIFT Overpressure Backlash & Overlap Differentiation From XIAN.TIME_VOID**:
     - Cap safe VOID_RIFT at 85; pushing 86..100 grants +22% armor-ignoring dimensional cleave (DIMENSIONAL_BLADE_REAPER) at the risk of minor spatial meridian tear (-6% HP backlash if interrupted by THUNDER / GRAVITY_ANCHOR—giving GRAVITY_DAO a natural rock-paper-scissors counter against VOID_SPACE!).

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5. [PATH-WEATHER-05] Path #19: CELESTIAL_WEATHER (วิถีเทวอากาศและฤดูกาลสวรรค์ — 6 Subpaths)

- **Subpaths**: THUNDER_CLOUD_CALLER (alias THUNDERSTORM_CALLER), MONSOON_RAIN_WEAVER, BLIZZARD_FROST_HERALD, TYPHOON_GALE_SOVEREIGN, MIRAGE_FOG_SHROUDER, SOLAR_ZENITH_INVOKER.
- **Verified Code & Context Defects**:
  1. **Subpath ID Alias Drift (THUNDER_CLOUD_CALLER vs THUNDERSTORM_CALLER vs THUNDER_TEMPEST_LORD)**:
     - v200 test line 130 uses 'THUNDERSTORM_CALLER', 01 catalog lists 'THUNDER_CLOUD_CALLER', and visitors_clues.js checks

`preferredSubs` on the Tempest Fan relic! Register bidirectional aliases in `CULT.SUB_ALIASES` and include both in `preferredSubs`!

2. **Three Uncoordinated Weather Authorities (`W.weather` vs `W.sect.activeWeatherMelody` vs `SECT.FORMATION_WARFARE`)**:
    - `harmonizeSkyWeather` writes `W.sect.activeWeatherMelody.weatherKind`, while `MANTRA_SYSTEM` (`v550`), `QINGGONG` (`v540`), and `ai.js` check `W.weather.kind`! Unless `activeWeatherMelody` synchronizes/attunes local battlefield/sect weather perception (without breaking global climate cycles), `QINGGONG` and `MANTRA_SYSTEM` don't even feel the Weather Cultivator's melody!
3. **Zero Situational NPC Weather Attunement & Zero Off-Weather Weakness**:
    - Replace the blind `SIM.monthly` call with intelligent situational attunement:
        - **Peaceful Farming Season**: `MONSOON_RAIN_WEAVER` attunes gentle `'rain'` near herb gardens, boosting `A.farm` & `SPIRIT_HUSBANDRY` growth by `+15%`!
        - **Meditation Peak Hours**: `SOLAR_ZENITH_INVOKER` / `MIRAGE_FOG_SHROUDER` attunes `'clear'` / `'spirit_aurora'`!
        - **Incoming Sect Raid**: `THUNDER_CLOUD_CALLER` / `BLIZZARD_FROST_HERALD` attunes `'thunder_tempest'` / `'snow'` (`+18%` `SKY_QI` generation, slows enemy raiders!), while suffering `-10%` cast efficiency when forced to fight in opposing weather (e.g., `BLIZZARD_FROST_HERALD` inside `volcano` / `crimson_sun`)!

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6. `[PATH-GEOMANCY-06]` Path #22: `DRAGON_VEIN_GEOMANCY` (วิถีมังกรธรณีและฮวงจุ้ยภูมิลักษณ์ — 6 Subpaths)

- **Subpaths**: `LUOPAN_GEOMANCER_SAGE`, `MOUNTAIN_SPINE_ANCHOR`, `LEYLINE_WARD_ARCHITECT`, `EARTH_DRAGON_AWAKENER`, `JADE_FAULT_CRUSHER`, `SANCTUARY_PEAK_CUSTODIAN`.
- **Verified Code & Context Defects**:
    1. **Closed-Loop Stone Leak Verification & Three Disconnected Leyline Engines**:
        - Verify `CULT.GRAND_DAO_MECHANICS.anchorDragonVeinNode(W, master, peakId, 120)` routes its `120`-stone cost through **`ECO.pay(W, cost, ...)`** (so `W.world.resourcePool.stones` receives `+120` while `W.sect.stones` drops by `-120`, keeping `v200` line

152 `W.sect.stones === stonesBeforeLey - 120` passing AND preserving strict world stone conservation!).
    - Bridge `anchorDragonVeinNode` with `SECT.LEYLINE_GRID` (`v2.0.35`) and `SECT.FENG_SHUI` (`v2.0.31`) so when a `LUOPAN_GEOMANCER_SAGE` or `SANCTUARY_PEAK_CUSTODIAN` walks to a Sect Peak to anchor its Dragon Vein (`AI.ACTS.attuneDragonVein`), it boosts **both** that Peak's `SECT.PEAKS` barrier/cultivation aura and `SECT.ELEMENTAL_BARRIER`, with a realistic **30-day decay timer** requiring maintenance (`ความเสี่ยงสูงสุด: หากขาดการบำรุงรักษา ค่ายกลภูมิลักษณ์จะเสื่อมสลาย`)!
2. **Missing Home-Turf vs Away-Terrain Asymmetry**:
    - Geomancers should gain `+16%` DEF/MDEF & knockback immunity (`MOUNTAIN_SPINE_ANCHOR`) when fighting on anchored Sect turf or `mountain` regions, and `-8%` `VEIN_RESONANCE` gain when fighting on open water/ships (`island`/water routes)!
3. **Miswired Weapon Scaling in `COMBAT.strike` (`js/combat.js`)**:
    - Ensure `DRAGON_VEIN_GEOMANCY` scales with `ASTROLABE` (`หลัวผานธรณี`) + `ARRAY` proficiency rather than unpracticed fallback skills!

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7. `[PATH-GRAV-CHRONO-CHAOS-07]` Paths #23, #24 & #27: `GRAVITY_DAO` (6), `CHRONO_EPOCH` (6) & `PRIMORDIAL_CHAOS` (6)

7.1 `GRAVITY_DAO` (วิถีแรงโน้มถ่วงสวรรค์ — 6 Subpaths)

- **Subpaths**: `HEAVY_STAR_CRUSHER`, `ZERO_GRAVITY_DRIFTER`, `ORBITAL_REPULSION_WARDEN`, `SINGULARITY_WELL_CONTROLLER`, `METEOR_WEIGHT_STRIKER`, `TIDAL_LOCK_ARBITER`.
- **Critical Combat Bug (`gravityCrush` Ignored by Sect NPCs!)**:
    - `applyGravityField` sets `target.gravityCrush = 4` (with `bossImmune: true` on bosses—keep that exact contract!).
    - **However**, in `COMBAT.moveToward` (`js/combat.js:232–255`), `u.gravityCrush` and `ST.get(u).move` are **never checked in the `else` (`!u.foe`) branch**! In tournaments, duels, or against rival Gravity cultivators, sect NPCs walk/fly right out of `gravityCrush` at full speed!
    - Wire `gravityCrush > 0` in `moveToward` (both `foe` and `!foe`) to ground `aerialFlightActive`, block `isGapCloser`/`blink`, and apply `0.45×` movement speed while decrementing per tick!
- **Subpath Hooks**:

- ORBITAL_REPULSION_WARDEN: Deflect -22% incoming ranged LONG_BALLISTIC damage (BOW/NEEDLE/FOCUS projectiles).
- ZERO_GRAVITY_DRIFTER: -35% Qinggong Qi/stamina cost and +18 move / eva.
- GRAVITY_PRESSURE Overstrain: Exceeding 95 GRAVITY_PRESSURE while toughness < 45 inflicts self-bone strain (กดทับเส้นชีพจรตนเอง).

7.2 CHRONO_EPOCH (วิถีกาลเวลาและวัฏจักรยุคสมัย — 6 Subpaths)

- **Subpaths**: SUNDIAL_EPOCH_CHRONICLER, HOURGLASS_SAND_REAPER, REWIND_SPRING_HEALER, ACCELERATED_BLADE_FLASH, STASIS_SEAL_WARDEN, PAST_FUTURE_MIRROR_SAGE.
- **Distinct Temporal Mechanics to Wire**:
  - **Snapshot State Rewind (REWIND_SPRING_HEALER)**: Track rolling 3-tick HP snapshot u._chronoHpRing in COMBAT.tick; when TEMPORAL_SAND_REWIND / triggerTemporalRewind fires, restore 45% of HP lost in the last 3 ticks (capped at 28% MaxHP, 18-tick ICD) **and** rewind 1 active skill cooldown (rewoundCooldowns >= 1, preserving v200 assertion!).
  - **DoT Time-Acceleration (HOURGLASS_SAND_REAPER)**: On signature hits, advance existing burn/poison/bleed timers on target by 2 ticks immediately (dealing instant compressed erosion without creating free infinite duration!).

7.3 PRIMORDIAL_CHAOS (วิถีโกลาหลต้นกำเนิด — 6 Subpaths)

- **Subpaths**: HUN_DUN_ORIGIN_DEVOURER, YIN_YANG_MILLSTONE_SAGE, FIVE_PHASE_INVERSION_ADEPT, ENTROPY_FLAME_SOVEREIGN, GENESIS_LOTUS_CREATOR, ABYSSAL_CHAOS_HALBERD.
- **Missing Elemental-Fusion Catalyst & Five-Phase Clash Risk**:
  - Wire PRIMORDIAL_CHAOS (HUN_DUN_ORIGIN_DEVOURER / FIVE_PHASE_INVERSION_ADEPT) into COMBAT.COMPOSABLE_ENGINE.triggerElementalReaction: when CHAOS_ENTROPY >= 50, a Chaos strike acts as a **Wildcard Catalyst Element** capable of completing any primed 2E or 3E (and 4E-02 CHAOS_WUXING_GENESIS!) elemental reaction on the target!
  - Enforce supremeRisk: if CHAOS_ENTROPY >= 95 and meridianIntegrity < 55, trigger minor **Five-Phase Chaotic Backlash** (entropyDecay self-recoil), rewarding Chaos cultivators who temper BODY_REFINING / meridians!

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8. Implementation & Verification Checklist for GPT / Gemini (01E)

- [ ] **Fix Grand Dao Stat & Trait Parity (`js/stats.js`)**: Unify `compute` and `getBaseFast` onto `CULT.GRAND_DAO_MECHANICS.getActiveTraits(n)` with `curOrd = Math.max(trackOrd, Math.round(normalizedStage))` so Stage-8/Realm-9 `subpathTraits` unlock properly; scale Grand Dao Mantra bonuses by `PATHW[c.path]`; deduplicate `triggerTemporalRewind` & `transmuteChaosElement` between `cultivation.js` and `combat.js`.
- [ ] **Fix Typos & Closed-Loop Conservation**: Fix `'STAR_CHART_ORACLE'` $\to$ `'STAR_CHART_PROPHET'` in `ELEM.syncSpecialAccess`; alias `'THUNDER_CLOUD_CALLER'` $\leftrightarrow$ `'THUNDERSTORM_CALLER'`; route `anchorDragonVeinNode` stone cost through `ECO.pay`.
- [ ] **Wire Authentic Autonomous Loops Across All 8 Paths**:
  - `ASTRO_DIVINATION`: Night cultivation bonus, celestial omen/raid foresight, `BIG_DIPPER_LONGEVITY_SAGE` `+15% lifeMax`.
  - `BUDDHA_DHARMA`: `heartDemon` cleansing in `A.cultivate`, compassionate `yield` / Heart-Demon counseling merit vs wrathful anti-sin smite.
  - `VOID_SPACE`: `POCKET_REALM_ARCHITECT` `+45%` expedition `carryCapacity` + projectile absorption + `VOID_RIFT` overpressure risk.
  - `CELESTIAL_WEATHER`: Unify `activeWeatherMelody` with `W.weather` perception + contextual farm/meditation/raid weather attunement.
  - `DRAGON_VEIN_GEOMANCY`: Peak-anchoring `AI.ACTS` + 30-day upkeep decay + home-mountain vs water-route modifier.
  - `GRAVITY_DAO`: Enforce `gravityCrush` on `!u.foe` NPCs in `moveToward` + orbital ranged deflection.
  - `CHRONO_EPOCH`: 3-tick HP snapshot rewind + DoT time-compression.
  - `PRIMORDIAL_CHAOS`: Wildcard Elemental Reaction catalyst at `CHAOS_ENTROPY >= 50` + Five-Phase overload risk.